

Alexander Britton

Engineer | Client-Facing Deployment | AI & Data Systems | Technical Problem Solving

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Summary

Engineer who builds production systems and deploys them at client sites. Most energized bridging the gap between technical teams and the people who use what we build — scoping requirements, running implementations on-site, training end users, and staying on as the point of contact after handoff. Repeat founder and published researcher.

Experience

Driveline Baseball Tampa, FL
Forward Deployed Engineer Aug 2025 - Present

- Led **end-to-end technical deployment** at three university partner sites (FSU, OSU, LSU) — scoped requirements with coaching staff, installed hardware, configured software, delivered training, and provided ongoing remote support
- Built **Theia Bot** — a production Slack interface using Claude API + LangGraph that enables non-technical staff to control motion capture pipelines in natural language, eliminating dependency on engineering
- Developed **self-service database query tool** replacing manual SQL request workflows; non-technical staff now query production data independently, removing a recurring support bottleneck
- Co-led **org-wide technical enablement** — authored documentation, recorded tutorial videos, and ran onboarding sessions across engineering and coaching staff
- Designed **cost-optimized solution architecture** routing lightweight LLM tasks to smaller models, achieving ~10x cost reduction at equivalent quality

Driveline Baseball Tampa, FL
Sports Science Engineer Lead June 2024 - Aug 2025

- Built **8-camera motion capture frontend** (React/TypeScript) with session management, calibration UI, and API integration — designed for coaches and trainers with zero technical background
- Integrated **34,000+ biomechanics trials** into production MariaDB; designed ETL pipelines, data validation, and monitoring for ongoing ingestion
- Designed **two-stage ML pipeline** (XGBoost + neural network) for elbow torque prediction achieving $R^2=0.76$; annotated 1,500 training samples

CerebriTech Orlando, FL
Co-Founder & CEO 2021 - 2023

- Co-founded health-tech startup developing real-time concussion detection for athletes; led **customer discovery**, product roadmap, and technical development through StarterStudio accelerator
- **Top 5 Entrepreneur Under 25**, Orlando Business Journal (2022)

Projects

BeBrief — Pilot Briefing Application 2024 - Present
Aviation briefing app currently pursuing FAA approval. Led product design, technical development, and regulatory scoping from concept through active review.
[Website](#)

Self-Service Database Query Tool 2025
Production MCP server enabling non-technical staff to query MariaDB in natural language. Replaced manual SQL request workflows and eliminated a recurring engineering support bottleneck.

YOLOv8 Pitch Detection System 2025
Object detection model (mAP50 0.813) with CVAT annotation pipeline and trajectory scoring algorithm. End-to-end from data labeling to production deployment.

Motion Capture Frontend 2024
8-camera control UI (React/TypeScript) with session management, calibration, and iPad interface. Built for coaches and trainers with zero technical background.

RAG Agents & LLM Integrations 2024 - Present
Multi-step RAG agents using HuggingFace transformers and vector databases for autonomous document research. Custom MCP extensions and Claude Code skills for engineering workflows.
[GitHub](#)

Skills

Implementation & Enablement
On-site deployment, requirements scoping, client training, documentation, onboarding programs, remote support

AI & LLM Systems
Claude API, OpenAI API, LangGraph, RAG, MCP, prompt engineering, agentic workflows, coding agents

Data & Integration
SQL, MariaDB, REST APIs, ETL pipelines, data migration, system integration

ML & Computer Vision
PyTorch, YOLOv8, XGBoost, scikit-learn, HuggingFace

Languages & Tools
Python, TypeScript, React, SQL, MATLAB, Docker, Linux, Git

Education

Embry-Riddle Aeronautical University Bachelor of Science • 3.92 GPA, Summa Cum Laude
Mechanical Engineering Daytona Beach, FL • Aug 2020 - May 2024

Academic All-American | Division II Baseball (4-year starter)